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Certifying rare overflow risk of regional pickup-and-delivery plans

Quang-Vinh Dang

British University Vietnam, Hung Yen, Vietnam, vinh.dq4@buv.edu.vn, ORCID 0000-0002-3877-8024

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Abstract. Medical, cold-chain and dangerous-goods distribution often require that a routing plan overflow its vehicles with a probability below a service level between one in a thousand and one in a million. Checking such a level for a given plan is hard for simulation: crude sampling needs hundreds of millions of draws per route, and standard importance sampling can fail without warning. We study this verification problem for vehicle routing with simultaneous pickup and delivery, under gamma-distributed demands with correlated customers. We show that a route overflows through one of two modes: all deliveries at the start or all pickups at the end. A single importance-sampling shift misses the weaker mode and underestimates the risk by about eight tenths of that mode's computed probability mass; the error depends on the route's structure, not its region, and can hide inside the estimator's own variance. A mixture of the two computed modes removes the bias without pilot simulation, and it extends to route days with travel times and to hubs whose routes share demand factors. On 148 instances from Northern Vietnam and Guangdong (6,242 routes) the method agrees with a numerical reference within 0.6 percent and with crude Monte Carlo wherever that is affordable, and its confidence intervals reach nominal coverage where the single shift covers in one third of the cases. It is about fifteen times faster than the numerical reference at one percent precision. The practical finding is that plans built for a per-route risk of ten percent do not meet any of the levels at their planned capacity: hubs need 35 to 76 percent more capacity. We also examine quantum amplitude estimation and find that, on this problem, the structure of the failure, not a quadratic speed-up, is what makes certification cheap.

Key words: rare-event simulation, importance sampling, stochastic vehicle routing, chance constraints, quantum amplitude estimation

1. Introduction

Planning models for stochastic vehicle routing control risk through chance constraints, robust counterparts or distributionally robust ambiguity sets (Gounaris et al. 2013, Dinh et al. 2018, Ghosal and Wiesemann 2020); see Gendreau et al. (2016), Akhmetbek (2024) and Karimi et al.

(2026) for reviews, and Ghosal et al. (2024) and Wang et al. (2023), Yin and Zhao (2022) for recent risk-measure and distributionally robust formulations. They evaluate route risk inside the optimiser at moderate levels, with exact formulas, approximations or sampling. Applications with strict service levels need a step after optimisation: an estimate, with a confidence interval, showing that the plan actually produced meets $\mathbb{P}(\text{overflow}) \le \gamma$ for $\gamma \in [10^{-6}, 10^{-3}]$, for every route and for the plan as a whole, at the scale of a regional network.

At such levels crude Monte Carlo needs $1.96^2/(\varepsilon^2\gamma)$ draws for relative error ε at 95% confidence — 3.8×10^8 draws per route at $\gamma = 10^{-6}$ and $\varepsilon = 0.1$. Importance sampling (IS) is the standard remedy, and mean shifts tuned by the cross-entropy (CE) method are the practitioner's default (de Boer et al. 2005). The failure of a pickup-and-delivery route is a union of events — the vehicle can overflow after any stop — and single-mode IS can then fail without warning (Glasserman and Wang 1997). Mixtures over failure modes (Owen et al. 2019, Kurtz and Song 2013, Geyer et al. 2019) cure this when the modes are known, at the price of pilot simulation when they are not; adaptive and flow-based proposals for multimodal failure sets are an active line (Helal and Elvira 2026, Zhang and Ma 2026), but their fit error is not part of the reported interval. Figure 5 shows the pipeline we propose and Section 6 gives its steps.

Where IS is needed. For the demand-only route model used in planning there is a deterministic alternative: given the copula factors the demands are independent, so the overflow probability is a two-dimensional integral of a one-dimensional dynamic program. We implement it, call it the quadrature reference (it is discretised, extrapolated and truncated, so not exact), and use it both as the reference and as a competitor (Section 7.3). It needs no tuning, but at the grid that gives about 1% accuracy it takes about 38 seconds per route and level (one core, an idle machine), against 2.5 seconds for the mixture estimator at the same precision. It does not extend to route days with stochastic travel and service times and lateness, where the event couples loads and times; there, of the methods we compare, only the mixture estimators are accurate, and the demand-only model is the setting in which their construction can be checked against a deterministic answer.

Contributions. (i) We formulate the certification of a given plan at service levels 10^{-3}–10^{-6}, route-wise and hub-wise, as a post-optimisation step. (ii) We derive the logarithmic asymptotics of the peak load of a pickup-and-delivery route under gamma marginals and a one-factor Gaussian copula: two end modes with spectral scales (Theorem 1, proof sketch). (iii) We show when single-shift IS loses logarithmic efficiency (Theorem 2) and measure its finite-level bias on real routes: the asymptotic threshold ($\varrho > 1/4$) holds for every route of our plans, but material bias appears

only when the computed mass $\hat{m}$ of the weaker end mode is large; the error is about $0.8\,\hat{m}$, follows the end-mode structure when the pickups are rescaled, and is invisible in the reported variance at $\gamma \le 10^{-5}$. (iv) We give a pilot-free stage-mixture estimator (SMIS) at computed dominating points, which on our routes reduces to two end modes, its extension to a route-day simulator without exact solution (MDIS), and a hub estimator for routes with shared factors, with attribution of hub risk to routes and a calibrated capacity search. (v) We evaluate on 148 region-scale instances against a quadrature reference, crude Monte Carlo at the levels where it is affordable, Glasserman–Li two-step IS, conditional Monte Carlo, subset simulation and CE, and we report the plan-level result: at their planned capacity none of the 148 hubs meets 10^{-3} or any smaller level. The IS ingredients are known — dominating points (Sadowsky and Bucklew 1990), multiple-path failure of single shifts (Glasserman and Wang 1997), mixtures as subsolutions (Dupuis and Wang 2007), factor-copula two-step schemes (Glasserman and Li 2005, Bassamboo et al. 2008) — and we do not claim new IS theory. What is new is their form for the pickup-and-delivery peak, the measured failure regime, and a certification procedure for regional plans.

2. Related work

Risk in vehicle routing. Chance-constrained and robust models (Gounaris et al. 2013, Dinh et al. 2018, Ghosal and Wiesemann 2020) handle route risk inside the optimisation with exact formulas, moment bounds or scenario sampling (Gendreau et al. 2016). Hazardous-materials routing measures risk as accident probability times consequence along links (Erkut and Verter 1998). We found no work that certifies an already optimised routing plan at rare levels with IS, and none that treats the mid-route peak of the simultaneous pickup-and-delivery load (Dethloff 2001) as a rare event. The closest IS precedent in logistics is stockout simulation in production–inventory systems (Glasserman and Liu 1996). Recent planning work keeps adding uncertain demand to pickup-and-delivery variants (Hu and Tao 2025), recourse policies (Salavati-Khoshghalb et al. 2019), distributionally robust and Wasserstein formulations (Yin and Zhao 2022, Wang et al. 2023, Ghosal et al. 2024) and CVaR approximations of chance constraints (Zhang et al. 2021, Thiebaut and Pessoa 2022); plans are then checked by simulation (Wang et al. 2025), at levels where crude sampling is enough. Correlated demand is modelled through a latent factor (Florio et al. 2023), and stochastic benchmarks with congestion and accidents exist (Heakl et al. 2025); none certifies rare levels.

Rare-event simulation. Efficiency notions and dominating points are surveyed by Asmussen and Glynn (2007); Glasserman and Wang (1997) show that a single shift is inefficient when the

rare set has several ways of occurring; Dupuis and Wang (2007) build efficient schemes from subsolutions, of which mixtures are instances; Owen et al. (2019) bound the variance of a mixture that samples each event of a union conditionally; CE with Gaussian mixtures handles multimodal failure regions (Kurtz and Song 2013, Geyer et al. 2019, Papaioannou et al. 2019). For Gaussian factor copulas, Glasserman and Li (2005) combine a factor shift with conditional exponential twisting, and Bassamboo et al. (2008) treat extremal dependence; for quadratic losses of Gaussian inputs, Glasserman et al. (2000) use the same eigenstructure that gives our scales. Subset simulation (Au and Beck 2001) and sequential Monte Carlo splitting (Cérou et al. 2012) need no mode information; we use subset simulation as the model-free rival. Recent work extends mixtures and splitting to multimodal and high-dimensional failure sets (Helal and Elvira 2026, Zhang and Ma 2026), and to the maximum of a dependent process (Conroy and Olvera-Cravioto 2022); Gaussian and Student-t factor copulas still use shifted proposals (Fang et al. 2026). What these methods do not give is a computed, structure-based mixture for the peak of a pickup-and-delivery route. On reporting, the sample size that importance sampling needs grows with the divergence between target and proposal (Chatterjee and Diaconis 2018), the Pareto-tail diagnostic of the weights (Vehtari et al. 2024) detects heavy tails but not a missed mode, and naive intervals for rare probabilities can fail to cover (Bai and Lam 2024).

3. Problem

A route visits customers $1, \dots, L$. The vehicle leaves the depot with all deliveries $d_1, \dots, d_L$ and collects pickups $p_1, \dots, p_L$. The load after stop k is $L_k = D_k + P_k$ with $D_k = \sum_{i>k} d_i$ and $P_k = \sum_{i \le k} p_i$ ($k = 0, \dots, L$), the peak is $F = \max_k L_k$, and the route overflows a capacity c if $F > c$.

Demand model. Each d_i is gamma with coefficient of variation cv, shape $\kappa = 1/\mathrm{cv}^2$ and scale $\theta_i^d = \bar{d}_i/\kappa$, likewise p_i. Dependence is a one-factor Gaussian copula per demand type: $d_i = G_{d,i}^{-1}(\Phi(Z_{d,i}))$ with $Z_{d,i} = \sqrt{\rho}\, W_d + \sqrt{1-\rho}\, E_{d,i}$, and similarly for pickups with a factor W_p independent of W_d. The input $u = (W_d, W_p, E_d, E_p)$ is standard normal. All customers of a hub share the two factors; hubs have separate factors. The baseline is cv = 0.3 and $\rho = 0.6$; Appendix B varies both, the pickup-to-delivery ratio and the marginal family. The independence of W_d and W_p is load-bearing: it makes the two end blocks independent, which the theory below uses.

Definition 1 (Certificate). Given γ, a relative precision ε and a confidence $1 - \delta$ (we use $\delta = 0.05$), a certificate for a route or a hub is an unbiased estimate $\hat{p}$ of its overflow probability p whose central-limit confidence interval has half-width at most $\varepsilon \hat{p}$, together with the check

$\hat{p}(1+\varepsilon) \le \gamma$; for a hub, also the attribution of p to its routes. The interval is asymptotic; Appendix B reports its empirical coverage.

4. Structure of the overflow event

For an index set S with gamma scales θ_S let $C_S = \rho \mathbf{1}\mathbf{1}^\top + (1-\rho)I$ and

$$\lambda(S) = \lambda_{\max}\left(\Theta_S^{1/2} C_S \Theta_S^{1/2}\right),$$

the largest root of $1 = \rho \sum_{i\in S} \theta_i/(\lambda - (1-\rho)\theta_i)$. Write $\lambda_k^d = \lambda(\{i > k\})$ for the delivery block of stage k, $\lambda_k^p = \lambda(\{i \le k\})$ for its pickup block, $\lambda_k = \max(\lambda_k^d, \lambda_k^p)$, $\lambda^\star = \max(\lambda_0^d, \lambda_L^p)$ and $\varrho = \min(\lambda_0^d, \lambda_L^p)/\lambda^\star \in (0,1]$.

PROPOSITION 1. $\max(D_0, P_L) \le F \le D_0 + P_L$.

THEOREM 1 **(proof sketch)**. *As $c \to \infty$ with the route fixed and unbounded gamma marginals,* $\log \mathbb{P}(L_k > c) \sim -c/\lambda_k$ *and* $\log \mathbb{P}(F > c) \sim -c/\lambda^\star$.

The proof (Appendix A) is a sketch: the uniformity of the logarithmic remainder of the gamma quantile (Lemma A1) is not written out. Our simulator clips Φ at $1 - 10^{-15}$, so its demands are bounded and the theorem describes the unclipped model; the clipping changes each input's law on a set of probability 10^{-15}, so on a route with $2L \le 80$ inputs it moves an overflow probability by less than 10^{-13}, far below $\gamma \ge 10^{-6}$.

So, on the logarithmic scale, a route overflows at the start (delivery block) or at the end (pickup block). Middle stages have strictly smaller scales: the stage where the two partial blocks cross has $\lambda_k \le \lambda^\star/2$ on homogeneous routes. A stage k carries relative mass $\exp(-c(1/\lambda_k - 1/\lambda^\star)(1 + o(1)))$ (Proposition 2); with $c \approx \lambda^\star \log(1/\gamma)$ the weaker end mode carries $\gamma^{(1/\varrho - 1)(1+o(1))}$. This is qualitative guidance only:

REMARK 1 (PRE-ASYMPTOTIC REGIME). For a three-stop test route with cv = 0.3, the exact dominating-point energy is 16% of its asymptotic value $c/\lambda^\star$ at $P \approx 9 \times 10^{-4}$ and 35% at $P \approx 1.6 \times 10^{-9}$; the logarithmic corrections of the gamma quantile are large at practical levels. The estimators therefore use computed dominating points, and the finite-level mass of a mode is measured by $\exp(-(E_m - E_{\min}))$ with exact energies E_m (Section 5).

Figure 1 shows the computed mode energies against the stage for one Northern Vietnam and one Guangdong route.

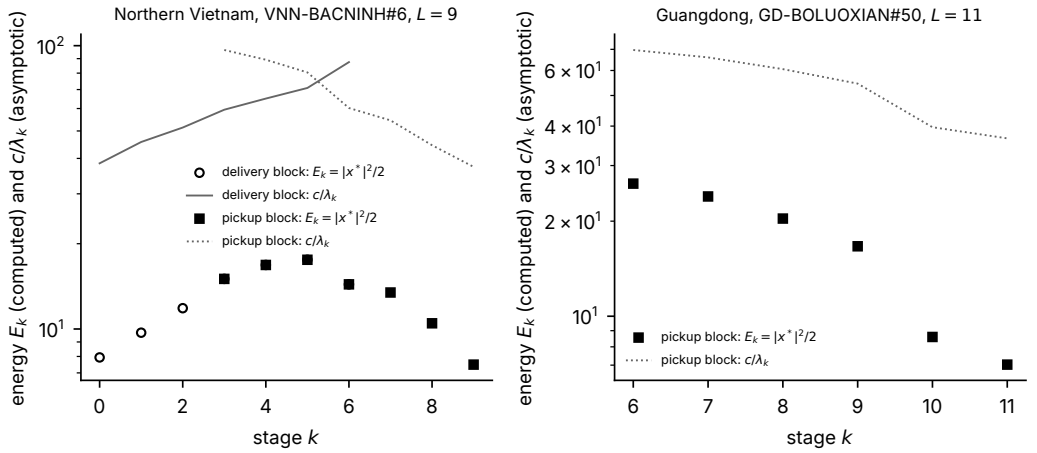


Figure 1 **Computed dominating-point energies E_k of the stage events $\{L_k > c\}$ (open circles: delivery blocks; filled squares: pickup blocks) against the stage k, with the asymptotic prediction c/λ_k (lines), at $\gamma = 10^{-4}$. Left: a route of Northern Vietnam (balanced pickups and deliveries), where both end modes exist. Right: a route of Guangdong, where pickups are 2.7 times the deliveries and no delivery block reaches c (the demands are bounded by the clipping of Φ): the failure has a single end mode. The computed energies are 5–10 times below the asymptote, so the figure shows the pre-asymptotic gap as well as the anatomy.**

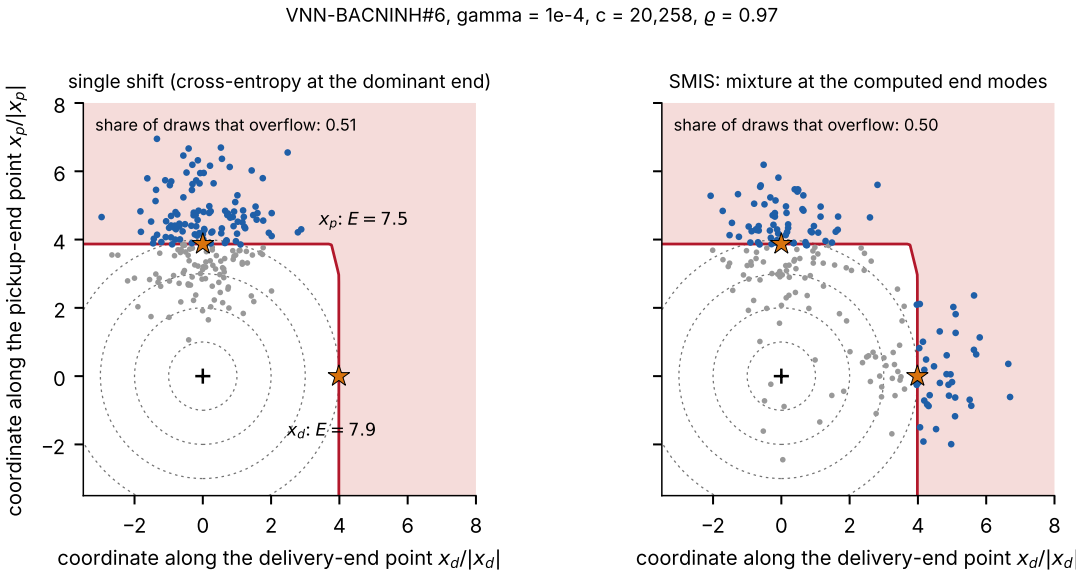


Figure 2 **The two end modes in the plane of the two dominating points (route VNN-BACNINH#6, $\gamma = 10^{-4}$). The shaded region is the overflow set in this slice through the origin; the stars are the delivery-end and pickup-end dominating points with their energies. Left: a single shift placed at one end puts draws only there, so the other mode is reached, if at all, with enormous weights. Right: the SMIS mixture places a component at each computed point.**

5. When a single shift fails

Mean-shift IS with shift τ uses the proposal $N(\tau, I)$ and weight $w(u) = \exp(-\langle u, \tau\rangle + |\tau|^2/2)$. Figure 2 shows why one shift is not enough on a balanced route.

THEOREM 2. *Let $\tau^\star$ be the asymptotic dominating point of the dominant end mode. (a) If $\varrho > 1/4$, the shift $\tau^\star$ is not logarithmically efficient:* $\log m_2(\tau^\star) - 2\log \mathbb{P}(F > c) \geq c\,(4/\lambda^\star - 1/(\varrho\lambda^\star))(1 + o(1)) > 0$. *(b) (proof sketch) If $\varrho \leq 1/4$ and the dominant end block has second eigenvalue at most $\lambda^\star/4$, the shift is logarithmically efficient.*

Part (a) rests on Theorem 1; (a) and (b) have different hypotheses, so $1/4$ is not a sharp threshold. The extra second moment in (a) comes from a region the proposal almost never visits, so the sample variance does not reveal it, and at finite levels the estimate is biased low by about the mass of the missed mode.

What the data show. The asymptotic criterion is uninformative for our plans: all 5 920 routes with two or more customers have $\varrho > 1/4$ (Guangdong median 0.36, Northern Vietnam median 0.93). The practical regime is narrower. On 120 routes (Table 2) the single CE shift has median error 0.14–0.23% in the ϱ bands with more than one route and $\varrho \leq 0.75$, close to the mixture estimators, and 2.9, 5.1, 6.0 and 6.1% (90% quantile 7.6, 21.7, 26.7, 39.9%) when $\varrho > 0.75$ at $\gamma = 10^{-3}, \ldots, 10^{-6}$. On those routes the error is an underestimate in 62% of the rows at 10^{-3} and in 93–95% at $\gamma \leq 10^{-4}$ (median signed error -4.3 to -4.7% at $\gamma \leq 10^{-4}$). It is visible at 10^{-3} and 10^{-4}, where the single shift's relative variance is 512 and 87, and invisible at 10^{-5} and 10^{-6}, where it is 5.2 and 5.5 (reported relative standard error 0.2% at 10^6 draws, against a 6% bias).

The missed-mode mass predicts the error. Let $\hat{m} = \exp(-(E_{\text{other}} - E_{\min}))$ be the weight that the computed dominating-point energies give the weaker end mode relative to the dominant one. It is computed from the route data in about a second and needs no sampling. We computed it for all 480 route–level rows (Figure 3, right). In the 320 rows with $\hat{m} < 10^{-6}$, all Guangdong, the single-shift error is at the noise level (median absolute error 0.19%, maximum 1.0%, against two standard errors of 0.43%); two more rows lie in $[10^{-6}, 10^{-3})$. In the 158 rows with $\hat{m} \geq 10^{-3}$, all from Northern Vietnam, the error is negative in 81–93% of the rows and grows with $\hat{m}$: the median is -0.4% for $\hat{m} \in [10^{-3}, 10^{-2})$, -2.2% for $[10^{-2}, 0.1)$ and -9.9% for $\hat{m} \geq 0.1$, and the median ratio of the absolute error to $\hat{m}$ is 0.98, 0.81 and 0.58. By level it is 0.22 at 10^{-3}, where the single shift still reaches part of the missed mode, and 0.74–0.79 at 10^{-4} to 10^{-6}. So at practical levels the single shift misses about $0.8\,\hat{m}$ of the probability, and $\hat{m}$ is a usable a-priori diagnostic: it is large exactly on the routes with $\varrho > 0.75$.

Region, ϱ and the assumed pickup rate. In our instances all rows with $\hat{m} \geq 10^{-3}$ are Vietnamese, where the pickup rate was set equal to the delivery rate because no regional pickup statistic is published, so the table above cannot separate the end-mode structure from region and assumption. We therefore changed the structure while keeping the routes. We rescaled the pickups of 20 Vietnamese routes by 0.5 and by 2 (Figure 4): ϱ falls from 0.93 to 0.49–0.51, the median $\hat{m}$ to below 10^{-10}, and the single-shift error is at most 1.7% in all 80 rows (median 0.1–0.2%), against medians of 5.1% and 6.1% at 10^{-4} and 10^{-6} over the 40 Vietnamese routes unscaled. In the other direction we halved the pickups of 30 Guangdong routes at 10^{-5} and 10^{-6}: ϱ rises to a median of 0.73, 42 of the 60 rows have $\varrho > 0.7$ and 13 have $\hat{m} \geq 10^{-3}$, four of those have an error beyond 2% (two routes, −41 to −43% and −6 to −8%), and the median ratio of error to $\hat{m}$ on the 13 rows is 0.84. The failure follows the end-mode structure, not the region. Correlating the delivery and pickup factors ($r = \pm 0.5$ between W_d and W_p, at 10^{-4}, 10 Vietnamese and 10 Guangdong routes each) leaves the Guangdong routes unaffected ($\varrho \approx 0.35$, errors below 0.9%) and makes the Vietnamese failure worse and, for $r = -0.5$, silent: the single shift underestimates by 2 to 80% (median 24%) on all 10 routes, while its reported relative variance is 4.3–5.3; for $r = +0.5$ the error has both signs and the relative variance is 5 to 25 000. The CE mixture agrees with the SMIS reference within 1.1% in both cases, so the reference is not the problem. SMIS keeps its construction here, since its dominating points are computed on the simulator, but the theory of Section 4 no longer applies and we report only this agreement. ϱ is computed in closed form and the threshold 0.75 remains empirical; $\hat{m}$ is the sharper indicator.

6. Estimators

SMIS. Algorithm 1 computes, for every stage k and block, a minimum-norm point u_m of $\{L_k(u) \geq c\}$ by constrained local minimisation (SLSQP) started on the block's Perron direction, with the exact quantile transform. The set is not convex, so u_m is a computed local minimiser, not a certified global one. Modes with energy $E_m = |u_m|^2/2$ within $\log(1/\eta)$ of the smallest are kept, and the estimator samples from

$$q = a_0\, N(0, I) + \sum_m a_m\, N(u_m, I), \qquad a_m \propto e^{-(E_m - E_{\min})},$$

with likelihood ratio $\varphi(u)/q(u) = \left(a_0 + \sum_m a_m e^{\langle u, u_m\rangle - |u_m|^2/2}\right)^{-1}$. The tuning constants are $\eta = 10^{-3}$ (hubs 10^{-4}) and $a_0 = 0.05$, fixed for every experiment; there is no pilot simulation. An optional variant (SMIS3) adds 3 points on arcs between modes of adjacent stages; it changes nothing

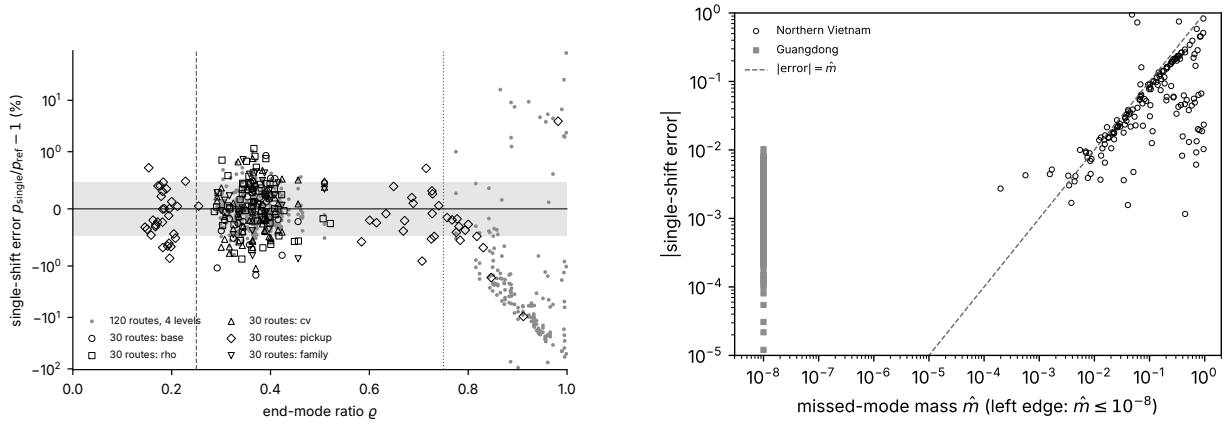


Figure 3 **Left: signed relative error of the single CE shift against ϱ (grey dots: 120 routes at four levels; open markers: 30 Guangdong routes under the sensitivity settings at $\gamma = 10^{-4}$); symmetric-log scale, grey band ± 2 median standard errors; dashed and dotted lines at $\varrho = 1/4$ and 0.75. Right: absolute error of the single shift against the computed mass $\hat{m} = \exp(-(E_{\text{other}} - E_{\min}))$ of the weaker end mode, for all 480 route–level rows (circles: Northern Vietnam; squares: Guangdong, all at $\hat{m} \le 10^{-8}$, drawn at the left edge), with the line $|\text{error}| = \hat{m}$.**

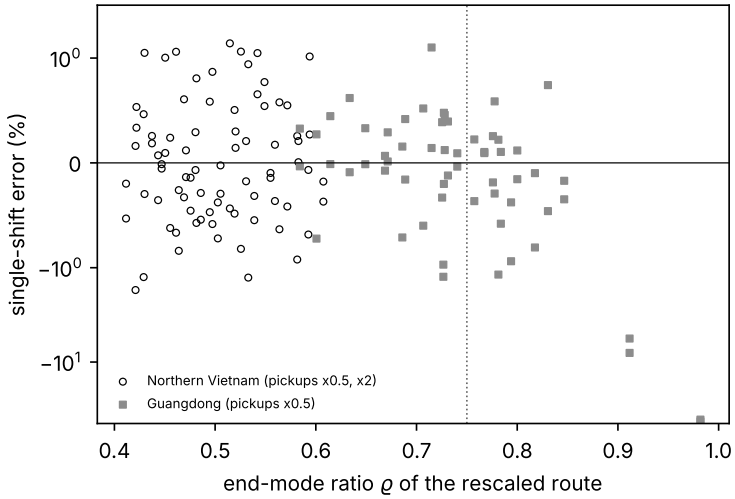


Figure 4 **Single-shift error against the end-mode ratio ϱ of the rescaled route. Open circles: 20 Vietnamese routes with the pickups scaled by 0.5 and by 2, at 10^{-4} and 10^{-6} (80 rows; $\varrho \approx 0.5$). Squares: 30 Guangdong routes with the pickups scaled by 0.5, at 10^{-5} and 10^{-6} (60 rows). Reference: SMIS with 10^6 draws; symmetric-log scale; dotted line at $\varrho = 0.75$.**

measurable on our routes and we report $n_{\text{arc}} = 0$ (SMIS0) throughout. Theorem 3 (Appendix) is a sketch of logarithmic efficiency under spectral-gap conditions; at $\rho = 0.6$ they hold for every end block of our plans, at $\rho = 0.3$ for 0.4% of the routes (1.4% of the end blocks), and Appendix B tests SMIS in both cases.

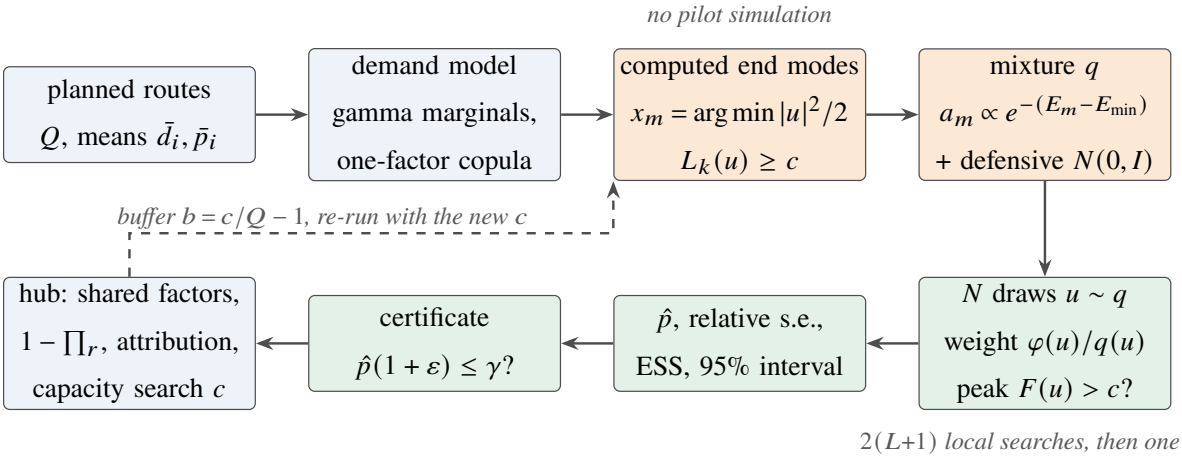


Figure 5 **The certification pipeline. Everything left of the draws is computed from the route and the threshold c, with no pilot simulation; the interval and the diagnostics come from the weighted draws. For a hub the same mixture is built per route and the capacity c is searched; the dashed arrow is that search.**

Algorithm 1 SMIS for one route

Require: route means $\bar{d}, \bar{p}$; cv, ρ; capacity c; η, a_0, N

1: **for** $k = 0, \ldots, L$ and block $b \in \{d, p\}$ **do**
2: $v \leftarrow$ Perron vector of $\Theta_b^{1/2} C \Theta_b^{1/2}$ on the block's inputs
3: $u_{k,b} \leftarrow \arg\min\{|u|^2/2 : L_k(u) \geq c\}$ by SLSQP from the ray along v ▷ skip if c is unreachable
4: **end for**
5: keep modes with $E_m \leq E_{\min} + \log(1/\eta)$; $a_m \leftarrow (1 - a_0) e^{-(E_m - E_{\min})} / \sum_{m'} e^{-(E_{m'} - E_{\min})}$
6: draw $u^{(1)}, \ldots, u^{(N)} \sim q$; $\hat{p} \leftarrow N^{-1} \sum_j \mathbf{1}\{F(u^{(j)}) > c\} \varphi(u^{(j)})/q(u^{(j)})$
7: **return** $\hat{p}$, its standard error, ESS

Cost: $2(L+1)$ local minimisations of dimension $2 + 2L$, each a few dozen quantile evaluations, then N simulator calls and $N \times$ (#modes) inner products.

CE baselines. The single CE shift uses the multilevel cross-entropy method with tail levels 0.1^j, 20 000 draws per iteration and at most 60 iterations per level. The CE mixture runs it for every stage event $\{L_k > c\}$ separately, estimates each stage probability with 20 000 draws under its shift, keeps the stages whose probability is at least 10^{-3} of the largest (at most 12), weights them by these probabilities and adds a defensive $N(0, I)$ component of weight 0.05. The draws spent on the ladders and on the stage probabilities (its pilot) are counted in its sample cost; with these settings they are 1.5–4.2×10^6 draws per route (measured on six routes, Table 5). In the 120-route study

(Tables 2 and 3) the classical rival, called the CE rival there, is the better of the two CE estimators on each route and level: the single shift (pilot 0.8–1.2×10^5 draws) when its estimate agrees with that of the mixture within two standard errors and has the smaller relative variance, and the mixture otherwise. This is the choice a practitioner with both estimators at hand would make; it makes the rival strong on routes where the single shift is unbiased and costly on the others.

MDIS. The same construction applies to any simulator that exposes its failure modes as scalar functions $g_m(u) > 0$ of a Gaussian input. We use it for a route-day model (Appendix C): demands as above, lognormal travel times per leg $T_l = \bar{t}_l \exp(\sigma_T(\sqrt{\rho_T}V + \sqrt{1-\rho_T}H_l) - \sigma_T^2/2)$ with $\bar{t}_l$ the road distance at 40 km/h, $\sigma_T = 0.25$ and a shared congestion factor ($\rho_T = 0.5$), service times of 10 minutes plus 6 minutes per tonne handled, and a failure if the vehicle overflows at any stage, arrives at any stop later than its nominal time plus a slack s, or returns later than nominal plus $2s$. Dominating points of the load, lateness and return modes are computed from several structured starts. No exact method applies: the event couples a maximum of copula sums with lognormal sums and load-dependent service.

Hubs. The routes of a hub share the factors, so the hub event $\bigcup_r \{F_r > c\}$ is again a union of stage events, now in dimension $2 + 2n$ for n customers. Every route's SMIS modes are embedded in the hub input (the two factors and that route's coordinates, zeros elsewhere), modes are kept over all routes with $\eta = 10^{-4}$, and the likelihood ratio

$$\log \frac{\varphi(x)}{q(x)} = -\log\left(a_0 + \sum_m a_m \exp\left(\langle x, \theta_m\rangle - |\theta_m|^2/2\right)\right)$$

is computed with sparse inner products, since θ_m has $2 + 2L_r$ nonzero coordinates. From the same draws we estimate each route's overflow probability and the probability that it is the only route to overflow — an attribution of hub risk to routes. A route that cannot reach c contributes no mode and zero probability.

Capacity calibration. To report the buffer a hub needs for γ, we search for c with $\hat{P}_{\text{hub}}(c) = \gamma$: a Laplace-type proxy $\log \sum_r e^{-E_r(c)} = \log \gamma$ gives a starting point c_0 (the proxy is not a bound: in a first run it set capacities at which the hub probability was 15 to 28 times below γ, at the 90% quantile and in median), then at most three secant steps on $\log \hat{P}_{\text{hub}}(c) - \log \gamma$ with 2×10^4 hub draws per evaluation, stopping when $|\log(\hat{P}_{\text{hub}}/\gamma)| < \log 1.25$; the final $\hat{P}_{\text{hub}}$ uses 10^5 draws. The buffer is $b = c/Q - 1$.

Table 1 Instances and plans (sample-average gate). Utilisation is the larger of mean total delivery and mean total pickup over Q, per route.

region	instances	customers	routes	stops q25/50/75 (max)	Q (t)	pickup/delivery	utilisation (median)	ϱ (median)
Northern Vietnam	26	2 612	375	6 / 7 / 8 (15)	15	1.01	0.62	0.93
Guangdong	122	37 487	5 867	5 / 6 / 8 (40)	30	2.70	0.65	0.36

Quadrature reference. Given the two factors (W_d, W_p) the demands are independent, so $\mathbb{P}(F > c \mid W)$ is a one-dimensional dynamic program over the discretised load, and $\mathbb{P}(F > c)$ is a two-dimensional trapezoid integral over the factors, with Richardson extrapolation over the number of load levels; for a hub the routes are conditionally independent. The details are in Appendix F.

7. Experiments

7.1. Instances and plans

We built 148 instances, one per depot area: the 25 provinces of Northern Vietnam (Hanoi split into two hubs) and 122 county-level areas of Guangdong (Table 1). Service points aggregate Kontur population hexagons (15 000 people per point in Vietnam, 4 000 in Guangdong) by weighted k-means; distances are OSRM road distances. Daily demand rates come from official postal statistics: 0.087 parcels per person per day in Vietnam (3.2 billion items in 2024) and, for Guangdong, 0.34 received and 0.91 sent per person per day (2024 volumes). The following are assumptions: 1 kg per parcel; truck capacities of 15 (Vietnam) and 30 (Guangdong) tonnes; the Vietnamese pickup rate equal to the delivery rate (no regional figure is published). The service points represent 35.3 million people in Northern Vietnam and 115.5 million in Guangdong (Kontur 2023 grid; the official Guangdong resident count for 2024 is 127.8 million, the difference being coastal hexagons outside the county polygons); Vietnamese provinces follow the pre-2025 boundaries. Plans are solved by ALNS with the sample-average chance gate; we certify these plans. Routes are short: the median route has 6 (Guangdong) and 7 (Vietnam) stops.

7.2. Routes under the demand model

The 120 routes are a random sample of the routes of the region plans (fixed seed), stratified by region and by tercile of the number of stops: 80 from Guangdong, 40 from Northern Vietnam, 4 to 15 stops, median 7. They are listed in round-robin order over the strata. The 30 routes of Section 7.3 are the first 30 in this order (15 per region), and the 10 routes of the coverage study are the first 10 of the 30 of Appendix B. The sensitivity study of Appendix B takes the first 30 rows of the route file, which lists the Guangdong routes first, so those 30 routes all belong to Guangdong; this is an artefact of the file order, not a design choice, and Section 5 tests the Vietnamese routes separately.

Table 2 Route model: relative error against the reference (quadrature reference where available, else CE rival), median / 90% quantile, and relative variance (median). 120 routes, $N = 10^6$; band $(0.5, 0.75]$ has one route and is omitted. The reference is the quadrature reference for 178 of the 480 route–level rows (routes with at most eight stops at 10^{-3} and 10^{-4}) and the CE rival for the other 302, so the CE columns at 10^{-5} and 10^{-6} are the reference itself ("ref."). Section 7.3 adds an independent reference at 10^{-4} and 10^{-6} for 30 routes.

γ	ϱ band	n	SMIS	CE rival	single shift	RV SMIS	RV single
10^{-3}	$[0, 1/3]$	17	0.19 / 0.58%	0.12 / 0.39%	0.17 / 0.45%	3.76	3.51
	$(1/3, 0.5]$	62	0.13 / 0.38%	0.09 / 0.30%	0.18 / 0.43%	3.78	3.51
	$(0.75, 1]$	40	0.22 / 0.44%	0.05 / 0.32%	2.86 / 7.57%	4.05	512
10^{-4}	$[0, 1/3]$	17	0.16 / 0.51%	0.17 / 0.52%	0.14 / 0.65%	4.50	4.23
	$(1/3, 0.5]$	62	0.17 / 0.45%	0.10 / 0.36%	0.18 / 0.44%	4.52	4.23
	$(0.75, 1]$	40	0.20 / 0.42%	0.07 / 0.36%	5.10 / 21.7%	4.71	87
10^{-5}	$[0, 1/3]$	17	0.19 / 0.55%	ref.	0.15 / 0.45%	5.18	4.88
	$(1/3, 0.5]$	62	0.22 / 0.45%	ref.	0.18 / 0.51%	5.18	4.88
	$(0.75, 1]$	40	0.26 / 0.58%	ref.	6.00 / 26.7%	5.31	5.22
10^{-6}	$[0, 1/3]$	17	0.24 / 0.61%	ref.	0.19 / 0.60%	5.79	5.46
	$(1/3, 0.5]$	62	0.26 / 0.55%	ref.	0.23 / 0.54%	5.79	5.46
	$(0.75, 1]$	40	0.22 / 0.52%	ref.	6.05 / 39.9%	5.88	5.50

We use four levels $\gamma \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$; c is set so that the overflow probability is close to γ. Every estimator uses $N = 10^6$ draws. The CE rival uses a pilot of 0.8–1.2×10^5 draws on the routes where it is the single shift and up to 3.9×10^6 where it is the mixture. Table 2 reports errors against the reference.

Against the quadrature reference. On the 89 routes with $L \le 8$ at $\gamma \in \{10^{-3}, 10^{-4}\}$ (178 cases), SMIS deviates from the quadrature reference by 0.15 and 0.16% in median, 0.38 and 0.44% at the 90% quantile, and at most 0.82%; the deviation is within 1.96 reported standard errors in 90% and 87% of cases (the program has its own discretisation error). The reference also covers $\gamma = 10^{-6}$ and routes with more than eight stops (Section 7.3): on the ten longest of 30 routes ($L \ge 9$) SMIS deviates by 0.13% (10^{-4}) and 0.20% (10^{-6}) in median and at most 0.58%.

Across all levels. SMIS has relative error at most 0.61% at the 90% quantile in every ϱ band with more than one route, relative variance 3.8–5.9 (CE rival 3.8–6.0, single shift 3.5–5.5 where it is unbiased), 2–7 components, and a median build time of 0.39–0.86 s. At $\gamma \le 10^{-5}$ the reference for routes without an quadrature reference is the CE rival.

Samples. Counting CE's pilot, the CE rival needs 224–276 times more draws than SMIS for $\varepsilon = 0.1$ at 95% confidence and 3.2–3.8 times more for $\varepsilon = 0.01$: the advantage is the saved pilot, not a smaller variance.

Table 3 Rivals against the quadrature reference, 30 routes (median over routes; error also as maximum). RV is per unit of work; $t(\varepsilon)$ is the one-core wall time for relative precision ε at 95% confidence, build and pilot included, measured on a shared machine with several jobs running (indicative; Table 5 repeats the main comparison on an idle machine). The reference needs no sampling; its time is per evaluation.

	$\gamma = 10^{-4}$			$\gamma = 10^{-6}$		
method	error med / max	RV	$t(0.01)$ (s)	error med / max	RV	$t(0.01)$ (s)
quadrature reference (fine grid)	reference	—	297	reference	—	303
SMIS	0.11 / 0.51%	4.6	2.6	0.16 / 0.58%	5.8	3.3
CE rival	0.16 / 0.38%	4.6	4.6	0.29 / 0.85%	5.9	6.4
single CE shift	0.41 / 23.5%	4.3	2.5	0.36 / 94.7%	5.5	2.5
Glasserman–Li, crude inner	5.0 / 47.8%	29	15	7.1 / 48.9%	57	32
Glasserman–Li, shifted inner	3.1 / 47.1%	11	5.4	2.9 / 47.6%	13	6.6
Glasserman–Li, tuned shift	2.9 / 45.3%	83	41	4.1 / 82.7%	15	16
conditional Monte Carlo	2.6 / 93.7%	6.5	9 310	3.1 / 63.0%	1.4	2 190
subset simulation	8.1 / 38.1%	716	431	21.6 / 108%	4 430	1 800
routes with $\varrho > 0.75$ only (15): median error						
single CE shift	6.2%	212		9.7%	5.5	
Glasserman–Li, shifted inner	15.5%	12		7.4%	13	
conditional Monte Carlo	13.7%	57		13.2%	153	
SMIS	0.15%	4.7		0.15%	5.9	

7.3. Classical rivals and the quadrature reference

On 30 of the 120 routes (15 with $\varrho > 0.75$) at $\gamma = 10^{-4}$ and 10^{-6} we compare all methods with the quadrature reference (Table 3). The rivals are Glasserman–Li two-step IS (Glasserman and Li 2005) with the factor shift at the dominant point (crude inner sampling, inner exponential shift, and a shift tuned on a pilot grid), conditional Monte Carlo over the two factors with the inner dynamic program and a tuned factor shift, and subset simulation (Au and Beck 2001) with five replications of 2 000 draws per level. Costs are stated per unit of work (a simulator call, or one inner program for conditional Monte Carlo) and converted to the wall time needed for relative precision ε at 95% confidence, build and pilot included, on one core.

Four findings from Table 3. First, SMIS agrees with the quadrature reference within 0.58% on every route, including the 15 routes with $\varrho > 0.75$ at $\gamma = 10^{-6}$, which the comparison with the CE mixture in Section 7.2 could not check independently. Second, the factor-based rivals inherit the single-mode failure, because both shift the factors towards one dominating point. On routes with $\varrho > 0.75$, Glasserman–Li with a shifted inner step is off by 7–16% in median at 2×10^4 calls; Table 4 repeats this at 10^6 calls (the budget of the other estimators), where its sampling error falls to about 0.5% and the error stays at −15% (10^{-4}) and −6.6% (10^{-6}), so the error is bias. Conditional Monte Carlo's relative variance jumps from about 1 to 57–153 per inner program on the same

Table 4 Budget-fair comparison, 30 routes, $N = 10^6$ simulator evaluations for each sampling method: absolute relative error against the quadrature reference (median / maximum) and relative variance per draw (median). SMIS, the two-end mixture and the CE mixture are unbiased on all routes; the Glasserman–Li estimator (shifted inner step) is biased low on balanced routes.

	$\gamma = 10^{-4}$		$\gamma = 10^{-6}$	
method	balanced (15)	other (15)	balanced (15)	other (15)
SMIS	0.15 / 0.57% (4.7)	0.17 / 0.38% (4.5)	0.12 / 0.57% (6.0)	0.24 / 0.58% (5.8)
two end modes only	0.13 / 0.59% (4.6)	0.08 / 0.45% (4.5)	0.13 / 0.39% (5.9)	0.19 / 0.60% (5.8)
CE mixture, pilot 2 000	0.10 / 0.30% (5.0)	0.12 / 0.56% (4.8)	0.13 / 0.51% (6.3)	0.23 / 0.49% (6.1)
Glasserman–Li, shifted inner	15.0 / 31.2% (30)	0.27 / 0.92% (10.6)	6.6 / 48.5% (13.9)	0.24 / 0.70% (12.2)

Table 5 Wall-clock time (s) to reach relative precision 0.01 at 95% confidence, $\gamma = 10^{-4}$, on an idle machine (one core, one task at a time, load average 1.0–1.2): build time plus $1.96^2\,\mathrm{RV}/0.01^2$ draws at the measured per-draw time; the quadrature columns are the wall time of one evaluation at two grids (41 nodes per factor axis, $V = 128, 256$ and $V = 256, 512$ levels). Balanced routes first.

route	L	SMIS	single CE shift	CE mixture	Glasserman–Li	quadrature (128,256)	quadrature (256,512)
VNN-DIENBIEN#2	10	3.4	3.1	22.6	8.7	41	222
VNN-HAIDUONG#14	9	2.4	2.9	19.6	7.1	36	201
VNN-HAIPHONG#3	10	2.8	220.6	19.5	11.5	40	219
GD-BOLUOXIAN#50	11	2.7	3.2	53.7	7.6	45	245
GD-DONGGUANSHI-2#9	7	1.8	2.0	23.9	5.0	29	155
GD-DONGGUANSHI-4#48	6	1.6	1.6	20.1	3.8	24	133
median		2.5	3.0	21.4	7.3	38	210

routes, the same missed-mode effect seen through the variance. Third, subset simulation needs no mode information but has a relative variance of 700–4 400 per draw and needs 7–30 minutes for $\varepsilon = 0.01$. Fourth, the cost of the quadrature reference depends on its grid, and the grid we used as reference is finer than needed (next paragraph).

Budget-fair comparison. Table 4 gives every sampling method 10^6 simulator evaluations on the same 30 routes. Two further estimators separate what SMIS adds from what CE adds. The first keeps only the two end modes of the SMIS mixture (delivery block of stage 0, pickup block of stage L, with the same weights and defensive component); it is as accurate and as efficient as full SMIS (median error 0.08–0.19%, relative variance 4.5–5.9), so on these routes the modes at intermediate stages add nothing measurable and the gain over a single shift is the second end mode at its computed dominating point. The second is a CE mixture with a pilot of 2 000 draws per iteration instead of 20 000: its pilot falls from 1.5–4.2×10^6 to 1.2–4.0×10^5 draws (the ladder still runs for every stage) and its accuracy equals that of SMIS (0.10–0.23% in median). A pilot is therefore not intrinsic to CE, but it remains the dominant cost of the mixture: Table 5.

Grids and longer routes. At 41 nodes with $V = 128$ and 256 the quadrature reference differs from the finer grid by 0.01% in median and costs a third of the time; SMIS reaches the same 1% precision

about 15 times faster than it. On eight routes with 15 to 33 stops, against crude Monte Carlo, SMIS deviates by at most 1.5% and the single shift fails on the one balanced route (Appendix D); Figure 9 in Appendix G shows how the cost of crude Monte Carlo and of SMIS scales with the level.

7.4. Sensitivity and coverage

We vary one input at a time on 30 Guangdong routes at $\gamma = 10^{-4}$ (correlation ρ, the coefficient of variation, the pickup means, lognormal instead of gamma marginals; Appendix B). SMIS stays within its sampling error in every setting, including $\rho = 0.3$ and lognormal marginals, which the theory does not cover. The capacity needed for 10^{-4} moves by −21% and +50% when the coefficient of variation goes from 0.3 to 0.2 or 0.5, one to two orders of magnitude more than the estimation error, so a certificate is only as good as the demand model.

Coverage. We drew 300 independent replications of 10^4 draws each and counted how often the nominal 95% interval contains the reference. On 10 Guangdong routes at 10^{-4} (base setting, the SMIS proposal built once) the interval of SMIS covers in 93.0–95.7% of the replications (pooled 94.4%, no zero estimates) and that of the single shift, on these routes with $\varrho < 0.75$, in 94.0–98.0% (pooled 95.8%). The harder case is the balanced routes at the rarer levels. On 10 Vietnamese routes with at most eight stops (ϱ from 0.84 to 0.99), at 10^{-5} and 10^{-6}, the SMIS interval covers in 93.0–98.7% of the replications (pooled 94.9% and 95.3%; the proposal is built once, but the sample is independent in each replication) against the quadrature reference, whose own error estimate is 1.6% and 0.9% of p in median. The single shift, rebuilt in every replication so that its pilot variability is included, covers in only 34% and 36% of the replications (routes between 0.7% and 96%): it is biased by −5.2% and −8.2% in median, against a standard deviation of 13.6% across replications. At 10^4 draws the relative standard error of SMIS is about 2.3%, so the central-limit interval is adequate well below the sample sizes we use. We therefore keep the word certificate for the SMIS interval at the levels and on the route types tested (1e-4 on Guangdong routes; 10^{-5} and 10^{-6} on short balanced routes), and nowhere else.

7.5. Route days without an exact method

On 60 route days, with the slack and the capacity set so that the load modes and the time modes each carry about half of the failure probability, MDIS agrees with crude Monte Carlo at 10^{-3} (median deviation 0.76% against about 1% standard error; 10^7 draws) and, on ten routes, at 10^{-4} and 10^{-5} (median 0.43% and 2.6%, within about two standard errors of 10^8–2×10^8 draws). The single CE shift is off by 0.4 to 45% and underestimates on all but one route. MDIS is built in 3–4 s against 43–55 s for the CE pilot (Appendix C, Table 9).

Table 6 Plans at their planned capacity Q: share of routes with overflow probability at most γ (route SMIS), and the overflow probability of a hub, by region (148 hubs, 6 242 routes). No hub passes at any γ.

	median	median	share of routes with $P_{\text{route}}(Q) \le \gamma$			
region	$P_{\text{hub}}(Q)$	$P_{\text{route}}(Q)$	10^{-3}	10^{-4}	10^{-5}	10^{-6}
N. Vietnam (26 hubs, 375 routes)	7.9%	2.2%	11.5%	9.9%	9.6%	9.1%
Guangdong (122 hubs, 5 867 routes)	10.8%	2.4%	12.2%	8.6%	6.7%	5.8%
all	10.2%	2.4%	12.2%	8.7%	6.8%	6.0%

7.6. Hubs

Do the plans pass? The plans were built with the sample-average gate at a per-route failure target of 0.10. We evaluated all 148 hubs and 6 242 routes at the planned capacity Q (hub SMIS with 10^5 draws, route SMIS with 2×10^4 draws per route). No hub meets any of the four levels (Table 6): the overflow probability of a hub at Q has a median of 7.9% in Northern Vietnam and 10.8% in Guangdong (10th to 90th percentile 6.2–9.9% and 8.1–13.8%). The median route overflows with probability 2.4% (90th percentile 3.8%), consistent with the target of 0.10 being met comfortably per route, and the shares of routes that pass are 12.2% at 10^{-3}, 8.7% at 10^{-4}, 6.8% at 10^{-5} and 6.0% at 10^{-6}. Most of the routes that pass are routes that cannot overflow at all: 4.7% of the routes have a largest possible load below Q. So a plan built for a per-route risk of 0.10 does not meet a service level of 10^{-3} or below anywhere; the certification step is what shows this, and the buffers below say how much capacity it takes to meet it.

Capacity needed. We certify 66 hubs: all 26 of Northern Vietnam and 40 of Guangdong, one drawn at random from each of 40 equal-size groups of the 122 hubs ordered by number of customers (25 to 597 customers, 5 to 100 routes). For each hub and $\gamma \in \{10^{-3}, 10^{-5}\}$ we search the common capacity c at which the hub overflow probability equals γ (Section 6) and report the buffer $b = c/Q - 1$ over the planned capacity (Table 7). All 132 searches converged, 121 of them after one secant step and 11 after two. Every route that can reach c has an SMIS mode; routes whose largest possible peak (with the demands bounded by the clipping of Φ) is at most c contribute exactly zero, which is why the quadrature hub program reports fewer reachable routes than the hub has (for example 3 of 5 at Bac Kan). The final estimate lies within a factor 1.25 of γ in every case (median $P_h/\gamma = 0.92$ at 10^{-3} and 0.94 at 10^{-5}, range 0.80–1.13). The final hub estimate takes a median of 36 s on one core (build 13 s plus 10^5 draws, 16–447 components, median 80); the whole task, with calibration, route-level baselines and checks, takes 3.4–4.4 minutes in median and at

Table 7 Hubs: capacity buffer $b = c/Q - 1$ needed for hub overflow probability γ (median / 90% quantile over hubs), the buffer the Laplace-type proxy would set, and the region probability $1 - \prod_h (1 - P_h)$ over the certified hubs (Guangdong: the 40 sampled hubs only).

γ	region	hubs	buffer b	proxy buffer	P_{region}
10^{-3}	Northern Vietnam	26	35.1 / 37.7%	53.0%	0.024
	Guangdong	40	43.4 / 47.3%	68.5%	0.036
10^{-5}	Northern Vietnam	26	64.5 / 68.0%	81.5%	2.5×10^{-4}
	Guangdong	40	75.8 / 80.0%	96.8%	3.8×10^{-4}

most 21 minutes (Figure 10 in Appendix G shows the distribution of hub risk, the buffers and the concentration on a few routes).

The buffers are large because the plans were built with a sample-average gate at a moderate level and a coefficient of variation of 0.3 with strongly correlated demands ($\rho = 0.6$): moving from 10^{-3} to 10^{-5} costs another 30 points of capacity. A per-hub level of 10^{-3} gives, over the hubs we certified, a probability of 2.4% (26 hubs of Northern Vietnam) and 3.6% (40 of the 122 hubs of Guangdong) that some hub overflows on a given day; the same per-hub level over all 122 Guangdong hubs would give about 10%. A regional level γ therefore needs a per-hub level near γ divided by the number of hubs.

Checks. The hub estimate lies between $\max_r p_r$ and $\sum_r p_r$, with tolerance four combined standard errors, at all 132 points; the hub probability is 3.4–3.7 times the riskiest route's and 0.37–0.51 times the sum over routes (medians). The hub estimates of Table 7 use 10^5 draws (relative standard error about 0.7%). Against 2×10^6-draw crude Monte Carlo on the 20 smallest hubs at $\gamma = 10^{-3}$ (relative standard error 2.2–2.5%), 19 of 20 standardised differences were within ±2 (one was 2.84), but their mean was +0.59, and against the quadrature reference five of six hub estimates were above it. To see whether this was a bias we re-estimated seven hubs at the same capacity with 10^6 hub draws (relative standard error 0.21%) and compared them with crude Monte Carlo at 2×10^7 draws (0.71–0.78%; 12 to 42 minutes per hub) and, for the six smallest hubs, with the quadrature reference. The seven hubs have 25 to 102 customers and 5 to 14 routes; the largest, Thai Nguyen, has no quadrature value. The differences from crude Monte Carlo are −0.9 to +1.6% (mean +0.02%, median −0.05%; standardised differences −1.16 to 2.11, mean 0.04) and those from the quadrature reference −0.07 to +0.29%. So the hub estimate shows no bias at 10^6 draws; the earlier positive differences were within the sampling error of the 10^5-draw estimates (and the reference's own error estimate of 2.8–9% of p is a conservative bound, since the two agree within 0.3%). The quadrature program takes 13–31 minutes per hub and level, against 41–146 s for the 10^6-draw hub estimate,

and its cost grows linearly with the number of routes. We did not check hubs at 10^{-5} against an independent method: crude Monte Carlo at that level needs 10^9 draws per hub, and the buffers at 10^{-5} rest on SMIS alone, which agrees with every independent reference we could afford at the other levels.

Shared factors. Treating the routes of a hub as independent overstates the hub probability by a factor 2.7 in median (4.6 at the 90% quantile) at 10^{-3} and 2.0 (2.9) at 10^{-5}, so a certificate built from route-wise estimates would demand too much capacity. With the shared factors, 47% (at 10^{-3}) and 33% (at 10^{-5}) of the overflow days of a hub have two or more routes overflowing (medians over hubs): standby capacity should be planned for clusters. The attribution shows how concentrated the risk is: the overflow probability of the riskiest route is 28–29% of the hub probability, and that of the three riskiest routes together 66–79% (medians over hubs; route events overlap, so these are not shares that sum to one; relative to the sum of the route probabilities they are 11–14% and 30–34%). Reinforcing a few routes, not all, is what a planner would do with the certificate.

7.7. Quantum amplitude estimation

Quantum amplitude estimation (QAE) estimates a probability with error ε from $O(1/\varepsilon)$ oracle calls instead of $O(1/\varepsilon^2)$ samples (Brassard et al. 2002, Montanaro 2015), and phase-estimation-free variants avoid the controlled operations of the original (Grinko et al. 2021). In finance it has been applied to VaR, CVaR and option pricing (Woerner and Egger 2019, Egger et al. 2021). Whether a speed-up survives the cost of the oracle is the open question: resource estimates for fault-tolerant pricing are large (Chakrabarti et al. 2021), depth-limited schemes narrow the gap (Herbert 2022), end-to-end accounting changes the comparison (Dalzell et al. 2023), and quadratic speed-ups exist for non-linear and heavy-tailed quantities (Blanchet et al. 2025). Quantum work on routing itself addresses deterministic problems (Feld et al. 2019, Phillipson 2024). We examine the overflow of one route, where the event is a predicate of the demand vector.

The oracle. The peak-load event is a reversible circuit (Figure 7 in Appendix E). A state-preparation unitary P loads the $2L$ demands, each on b qubits, in superposition; U_{eval} adds the demands stage by stage, compares each load with c, and counts the stages above it; one objective qubit records whether the count is positive. The amplitude of that qubit is the overflow probability p. We built this circuit in Qiskit and counted its resources from the compiled circuit, not from a formula (Figure 8 in Appendix E): for $L = 8$ stops and $b = 8$ bits per demand the oracle has 995 Toffoli gates per call, and the count grows linearly in L and in b.

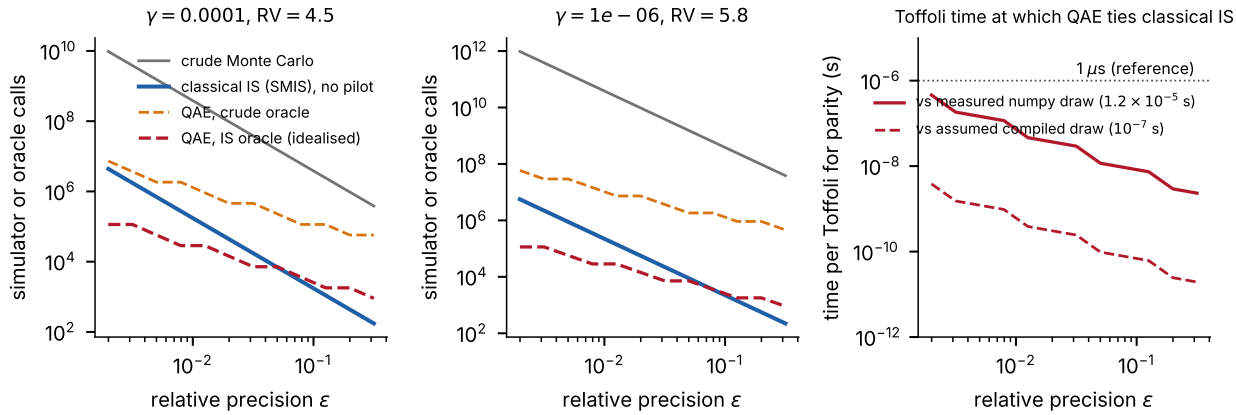


Figure 6 **Simulator or oracle calls to reach relative precision ε at two service levels (left, middle): crude Monte Carlo, classical IS with the SMIS relative variance, and QAE with a crude and with an idealised IS oracle. Right: time per Toffoli gate at which QAE with an IS oracle ties classical IS, against the measured numpy draw time and an assumed compiled draw time.**

Calls. On the same 120 routes, counting oracle calls only (no qubit or error-correction cost) with explicit constants for a rigorous schedule, QAE needs 0.003–0.5 times the calls of crude Monte Carlo, but 27–3 550 times more than the sampling stage of classical IS (2.5–380 times more if the IS pilot is counted). With an IS-equipped oracle it needs fewer calls than classical IS only in an idealised accounting at $\varepsilon \le 3\%$ (0.14–0.48 times). Figure 6 shows the calls as functions of ε; its last panel converts the comparison into the time per Toffoli gate at which QAE with an IS oracle ties classical IS. At $\varepsilon = 1\%$ that time is about 4×10^{-8} s, well below the 1 μs order commonly assumed for fault-tolerant logical gates; the 1 μs line is a reference value, not a hardware claim. On this problem the mode structure, not the quadratic speed-up, is what makes certification cheap.

8. Conclusion

Rare overflow of a pickup-and-delivery route has a simple anatomy on the logarithmic scale: two end modes with closed-form scales, and the computed mass of the weaker one predicts how much a single importance-sampling shift underestimates (about 0.8 times that mass), in both regions. A mixture at the two computed dominating points removes the bias without pilot simulation, extends to route days and to hubs with shared factors, and gives intervals that cover at their nominal level where the single shift covers in a third of the cases; it agrees with the quadrature reference and with crude Monte Carlo within their sampling error and is about 15 times faster than the quadrature reference at 1% precision. The plan-level result matters most for practice: plans built for a per-route risk of 0.10 do not meet any level between 10^{-3} and 10^{-6} at their planned capacity (a hub overflows

with probability about 8–11%); they need 35–43% more capacity at 10^{-3} and 64–76% at 10^{-5}, ignoring the shared factors would overstate hub risk two- to threefold, and demand variability moves the needed capacity far more than estimation error does. Open points are full proofs at the stated generality, region-wide factors that couple hubs, a theory for correlated pickup and delivery factors, a hub check against an independent method at 10^{-5}, and demand and travel models estimated from operator data.

Code and data. All code, instances, plans and scripts are in `https://github.com/vinhqdang/stochastic_vrp` (directory `svrpspd_wdro`; Appendix H).

Appendix A: Proofs and sketches

Proof of Proposition 1. $L_0 = D_0$ and $L_L = P_L$ are stage loads, and $D_k \le D_0$, $P_k \le P_L$ for all k. Q.E.D.

Lemma A1 (quantile; stated without proof). For the gamma law, $x(z) = G^{-1}(\Phi(z)) = \theta z^2/2 + \theta(2\kappa - 1)\log z + O(1)$ as $z \to \infty$, and $0 \le x(z) \le x(0)$ for $z \le 0$.

Proof sketch of Theorem 1. Upper bound: by Lemma A1, $D_k \le \frac{1}{2} Z^\top \Theta Z + R$ with R a sum of logarithmic terms; for $t < 1/\lambda$, $\mathbb{E}\exp(t\,\frac{1}{2} Z^\top \Theta Z) = \det(I - t\Theta^{1/2} C \Theta^{1/2})^{-1/2} < \infty$, so Chernoff's bound gives the rate $1/\lambda$, and R is negligible if Lemma A1 holds uniformly (not written out). The delivery and pickup blocks have disjoint inputs, so their moment generating functions multiply and L_k has rate $1/\lambda_k$. Lower bound: a ball of radius $O(1)$ around $(1+\eta)\sqrt{2c/\lambda}\,u^\star$, with $u^\star = B^\top \Theta^{1/2} v/\sqrt{\lambda}$ ($Z = Bu$, $BB^\top = C$, v the Perron vector), lies in the event. For F use Proposition 1.

Proposition 2 **(from Theorem 1)**. *For a fixed gap $\lambda_k < \lambda^\star$, $\mathbb{P}(L_k > c)/\mathbb{P}(F > c) = \exp(-c(1/\lambda_k - 1/\lambda^\star)(1 + o(1)))$.*

Lemma A2. Let $h(u) = \frac{1}{2} u^\top M u$ with $M = A^\top \Theta A$, $A \ge 0$, eigenvalues $\lambda_1 > \lambda_2$, $\kappa = \lambda_1/\lambda_2 \ge 4$, $h_+(u) = \frac{1}{2}|\Theta^{1/2}(Au)_+|^2$, and $\tau = r v_1$ with $r^2 = 2c/\lambda_1$. Then $h_+(u) \ge c$ implies $|u + \tau|^2 \ge 4|\tau|^2$. *Proof.* Write $u = a v_1 + u_\perp$, $b = |u_\perp|$. If $a \ge 0$, then $c \le \frac{1}{2}(\lambda_1 a^2 + \lambda_2 b^2)$, so $b^2 \ge \kappa(r^2 - a^2)$ and $|u + \tau|^2 \ge (1 - \kappa)a^2 + 2ar + (1 + \kappa)r^2$, a concave function of a whose minimum on $[0, r]$ is $\min((1+\kappa)r^2, 4r^2) = 4r^2$ for $\kappa \ge 3$; for $a \ge r$, $|u + \tau|^2 \ge (a + r)^2 \ge 4r^2$. If $a < 0$, then $A v_1 \ge 0$ gives $(Au)_+ \le (Au_\perp)_+$, so $b^2 \ge \kappa r^2 \ge 4r^2$. Q.E.D.

Proof of Theorem 2(a), given Theorem 1. $\tau^\star$ acts on the dominant block's inputs only and the other end event depends on the other block's inputs only, so $m_2 \ge \mathbb{E}[w]\,\mathbb{P}(\text{other end}) = e^{|\tau^\star|^2}\mathbb{P}(\text{other end}) = \exp(2c/\lambda^\star - c/\lambda_{\text{other}} + o(c))$, while $2\log\mathbb{P}(F > c) = -2c/\lambda^\star(1 + o(1))$. Q.E.D.

Sketch of Theorem 2(b). Proposition 1 and Lemma A2 applied to the block-diagonal form of the two end blocks; the logarithmic remainders would be handled by Cauchy–Schwarz.

Theorem 3 **(sketch)**. *With a constant defensive weight, SMIS with modes at the asymptotic dominating points is logarithmically efficient if every retained block has $\kappa \ge 4$ and in every retained stage one block dominates the other by a factor of at least 4; stages with $\lambda_k \le \lambda^\star/2$ need not be retained.*

Table 8 **Sensitivity, 30 Guangdong routes, $\gamma = 10^{-4}$, 2×10^5 draws: absolute relative error (median / 90% quantile), median ϱ, and the capacity needed for 10^{-4} relative to the base setting (median).**

setting	ϱ	SMIS	single shift	CE mixture	c/c_{base}
base ($\rho = 0.6$, cv 0.3, gamma)	0.37	0.27 / 0.77%	0.24 / 0.82%	0.26 / 0.82%	1
$\rho = 0.3$	0.37	0.34 / 0.78%	0.26 / 0.72%	0.28 / 0.60%	0.87
$\rho = 0.8$	0.37	0.33 / 0.95%	0.39 / 0.88%	0.39 / 0.85%	1.08
cv = 0.2	0.37	0.35 / 0.83%	0.26 / 0.67%	0.30 / 0.81%	0.79
cv = 0.5	0.37	0.37 / 0.66%	0.38 / 0.66%	0.35 / 0.66%	1.50
pickups ×0.5	0.73	0.32 / 0.81%	0.34 / 0.99%	0.43 / 0.89%	0.50
pickups ×2	0.18	0.40 / 0.77%	0.34 / 0.66%	0.28 / 0.61%	2.00
lognormal marginals	0.37	0.43 / 0.86%	0.29 / 0.67%	0.35 / 0.70%	1.09

Appendix B: Sensitivity to the demand model

On 30 Guangdong routes (4 to 14 stops) at $\gamma = 10^{-4}$ we vary one input at a time: $\rho \in \{0.3, 0.8\}$, cv $\in \{0.2, 0.5\}$, the pickup means scaled by $s \in \{0.5, 2\}$, and lognormal instead of gamma marginals with the same mean and cv (Table 8). In each setting the capacity is reset so that the overflow probability is close to 10^{-4}, every estimator uses 2×10^5 draws, and the reference is the quadrature reference for gamma marginals with $L \le 8$ (25 routes) and a 10^6-draw CE mixture otherwise.

SMIS stays within its sampling error (median 0.27–0.43% against a relative standard error of about 0.47%) in every setting, including $\rho = 0.3$, where the spectral-gap condition of Theorem 3 holds for only 0.4% of the routes of our plans (1.4% of the end blocks), and lognormal marginals, which the theory does not cover. The single shift is also accurate on these routes except where halving the pickups makes the two end scales nearly equal (Section 5). The last column is the model risk: the capacity needed for 10^{-4} moves by −21% and +50% when cv goes from 0.3 to 0.2 or 0.5, by −13% and +8% for ρ, and by +9% for lognormal marginals. These shifts are one to two orders of magnitude larger than the estimation error, so a certificate is only as good as the demand model, and cv in particular should be estimated from operator data.

Appendix C: Route days without an exact method

On 60 routes we set the slack s and the capacity so that the load modes and the time modes each carry about half of the failure probability — a design that makes the failure multimodal on purpose, to test the estimator where a single shift is known to struggle. The threshold search is common to all methods and not counted. MDIS keeps 6–8 modes (median) of the 62–65 candidates (median); 16–21 candidate modes per route cannot reach their threshold and are dropped. Table 9 gives the results. At $\gamma = 10^{-3}$, against 10^7-draw crude Monte Carlo (relative standard error about 1%), MDIS deviates by 0.76% in median and 1.77% at the 90% quantile, CE by 0.89% and 1.75%, and the single CE shift by 2.2% and 22%. At 10^{-4} and 10^{-5} we also ran crude Monte Carlo, with 10^8 draws on 7 routes (1.0–1.2% relative standard error) and 2×10^8 draws on 3 routes (2.2%). MDIS deviates from it by 0.43% in median (2.1% at most) at 10^{-4} and by 2.6% (4.9% at most) at 10^{-5}, within about two standard errors of the reference in every case; the CE mixture is as close; the single CE shift is off by 0.4 to 38% at 10^{-4} and by 12 to 45% at 10^{-5}, underestimating on every route but one. MDIS and the CE mixture, built independently from the same simulator, agree within 0.24% in median and 0.59% at the 90% quantile over all 60 routes. MDIS is built in 3–4 s against 43–55 s for the CE pilot, and CE needs about 1 800 times more draws for $\varepsilon = 0.1$ and 18–19 times more for $\varepsilon = 0.01$.

Table 9 Route-day simulator, 60 routes: deviation (median / 90% quantile), relative variance (median), build time (median).

γ	MDIS vs crude	CE vs crude	single vs crude	MDIS vs CE	MDIS vs single	RV MDIS / CE / single	build MDIS / CE (s)
10^{-3}	0.76 / 1.77%	0.89 / 1.75%	2.18 / 21.9%	0.20 / 0.47%	1.98 / 28.4%	4.33 / 4.41 / 742	3.1 / 43.1
10^{-4}	—	—	—	0.24 / 0.53%	3.42 / 63.3%	5.25 / 5.24 / 1430	3.5 / 49.4
10^{-5}	—	—	—	0.23 / 0.59%	9.25 / 97.6%	6.12 / 6.04 / 357	3.8 / 55.1

Appendix D: Quadrature grids and longer routes

The quadrature reference at coarser grids. The reference of Table 3 uses 41^2 factor nodes and $V = 256$ and 512 levels. On the same 60 route–level rows, 41 nodes with $V = 128$ and 256 differs from it by 0.01% in median (0.44% at the 90% quantile, 0.97% at most) and costs about a third of the time (95 s against 300 s per evaluation under load). With 21 nodes the error is 1.2% in median (8.5% at most) and with 11 nodes 48%: the trapezoid rule on $[-6, 9]^2$ with a step above 0.5 does not resolve the tails that dominate the rare event. The cheapest grid that keeps the error below 1% is therefore 41 nodes with $V = 128$ and 256, and at that grid SMIS reaches the same 1% precision about 15 times faster (12 to 17 times on the six routes of Table 5, median 2.5 s against 38 s) and 83 times faster than the finer grid used as reference (210 s). The Glasserman–Li estimator needs 7 s but is biased on balanced routes, the CE mixture 21 s (pilot included), and the single shift 3.0 s except on the balanced route where it fails, where its relative variance makes it need 221 s, as much as the quadrature reference. *Longer routes.* The 30 routes above have at most 13 stops. We also ran eight routes with 15 to 33 stops, drawn at random from eight equal groups of the routes of the plans with 15 or more stops (seven from Guangdong, one from Northern Vietnam). The quadrature reference is too expensive for these at the accuracy we need, so the reference is crude Monte Carlo: 5×10^7 draws at 10^{-3} (relative standard error 0.45%) and 10^8 at 10^{-4} (1.0%). SMIS deviates from it by 0.34% in median and 0.93% at most at 10^{-3}, and by 0.90% and 1.5% at 10^{-4}, at most about two standard errors of the reference; the CE mixture is as accurate. The single shift is accurate on the seven Guangdong routes (ϱ from 0.34 to 0.41) and fails on the one balanced Vietnamese route ($\varrho = 0.96$): +5.4% at 10^{-3} and −9.2% at 10^{-4}, with relative variances of 7 016 and 24 386 per draw, so on this route the failure shows in the variance. The 33-stop route behaves like the others. Eight routes are a small sample, and none has more than 33 stops.

Section 7.6 compares the hub estimate with the quadrature reference and with crude Monte Carlo.

Appendix E: The quantum oracle

Appendix F: Quadrature reference

Given (W_d, W_p) the demands are independent, so $\mathbb{P}(F > c \mid W)$ is a one-dimensional dynamic program over the discretised load (we use $V = 256$ and 512 levels with Richardson extrapolation), and $\mathbb{P}(F > c) = \iint \mathbb{P}(F > c \mid w)\, \varphi(w_d)\varphi(w_p)\, dw$ is a product trapezoid on $[-6, 9]^2$. For a hub, the routes are conditionally independent, so $\mathbb{P}_{\text{hub}} = \iint [1 - \prod_r (1 - \mathbb{P}(F_r > c \mid w))]\varphi\varphi\, dw$. The error estimate combines the Richardson difference and a nested-grid quadrature difference.

Appendix G: Additional figures

Appendix H: Code and data

The instances, plans, simulators, estimators and every script that produces a number, table or figure in this paper are in the repository `https://github.com/vinhqdang/stochastic_vrp` (directory `svrpspd_wdro`; commit `a68ed2a3f4`);

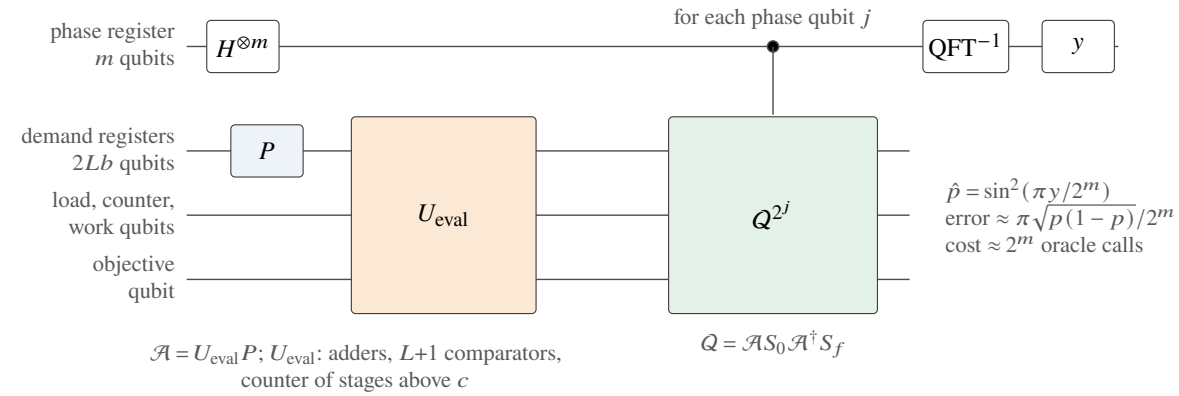


Figure 7 **Amplitude estimation around the peak-load oracle. The estimate is read from y; the cost is the number of applications of $\mathcal{A}$ and $\mathcal{A}^\dagger$ inside the controlled powers of Q.**

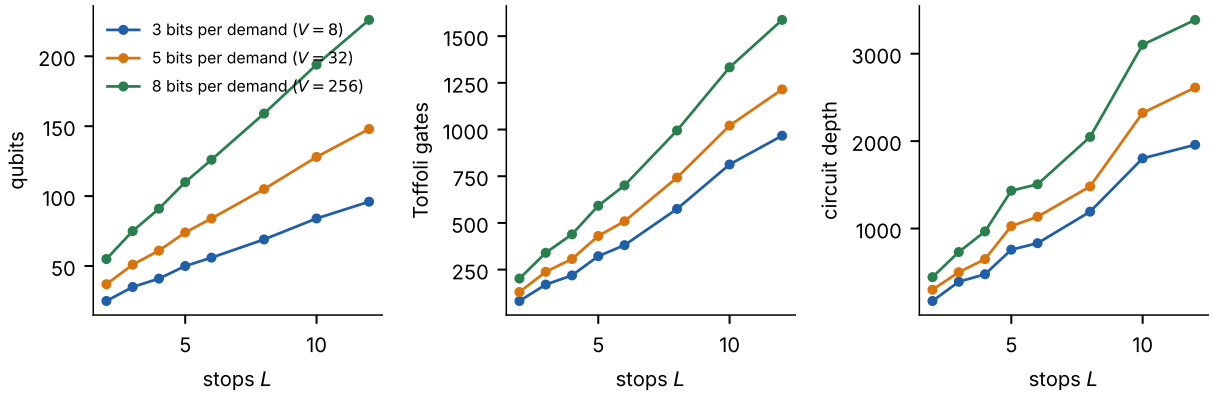


Figure 8 **Resources of the coherent peak-load oracle, from the compiled Qiskit circuit, against the number of stops L for several demand precisions.**

a README maps each table and figure to its script, and a copy will be deposited with a persistent identifier at acceptance. The long runs write per-task checkpoints and can be split into shards and merged.

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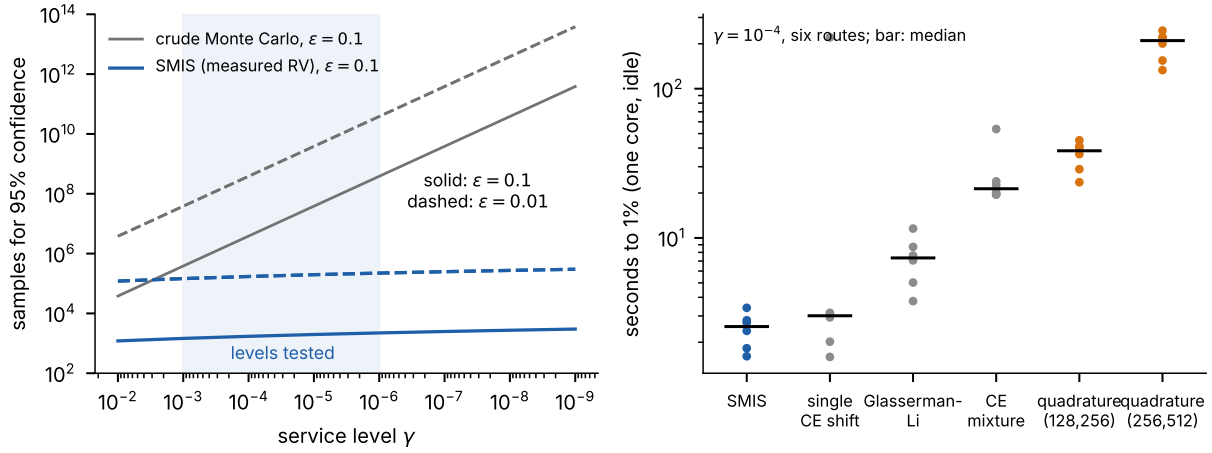


Figure 9 **Cost of certification. Left: draws for 95% confidence at relative precision ε, as a function of the service level γ, for crude Monte Carlo and for SMIS with its measured relative variance (solid $\varepsilon = 0.1$, dashed $\varepsilon = 0.01$). Right: seconds to 1% precision on the six routes of Table 5 (bar: median).**

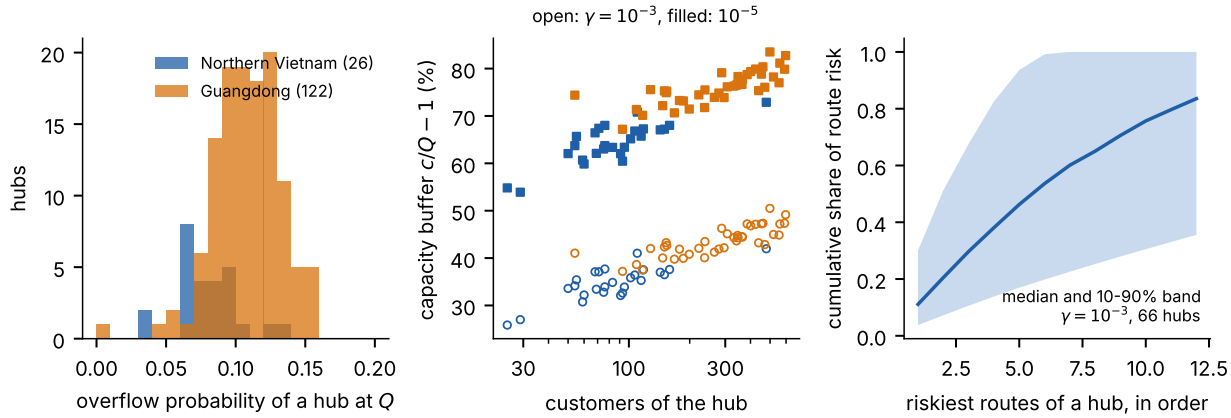


Figure 10 **Hubs. Left: overflow probability of each hub at its planned capacity Q (no hub is below 10^{-3}). Middle: capacity buffer needed, against the number of customers of the hub. Right: cumulative share of the route risk carried by the riskiest routes of a hub, in order (median and 10–90% band over the 66 hubs).**

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